

A Dual-Robot Cooperative Arc Welding Path Planning Algorithm based on Multi-Objective Optimization

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Abstract: In this paper a novel improved multi-objective genetic algorithm (R-NSGA-II) is proposed to solve the problem of dual-robot arc welding path planning. We strive to find a low-cost, fast and more efficient solution for large complex components. Firstly, a multi-objective optimization model of path planning is established by considering various variables and constraints in the actual welding process. Then, the R-NSGA-II introduces a recombination mechanism to replace the mutation operation of NSGA-II; this can increase the diversity of individuals and populations, and find the global optimal solution with greater probability. Finally, in order to verify feasibility and effectiveness of the proposed algorithm, it is used to plan some typical welding seams of a large complex component, the NSGA-II and OMOPSO are used for comparison. The simulation demonstrates that R-NSGA-II can obtain better Pareto front than the other two algorithms. Particularly, the R-NSGA-II reduces the waiting time by 42.39% and the no-load distance by 10.50% compared with NSGA-II.

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Keywords: Path planning, dual-robot, arc welding, multi-objective optimization, large complex components.

1. INTRODUCTION

In recent years, the welding of large and complex components has more and more extensive needs. However, manual welding has used by most factories, which has the disadvantages of low efficiency, poor welding quality and poor working environment, it is urgent to use robots to replace manual welding (Ding et al., 2015). Most single robots can only engage in simple and repeated tasks in a fixed working environment. In order to meet the requirements of task complexity and intelligent operation in unstructured environments, two or more robots show advantages in performing such tasks through mutual cooperation. Therefore, the dual robot system is gradually introduced into the welding process of large and complex components to improve the welding quality and production efficiency (Wang et al., 2021). In the robot system, the first problem to be solved is the task allocation (or path planning) of multiple robots, it is the premise for robots to perform all subsequent tasks. At present, the off-line planning based on manual is mostly used to realize the path planning of dual-robot in the production field (Feng et al., 2020). However, large complex components have a great quantity of welding seams, and there are many complex constraints between welding seams, such as synchronous welding constraints, welding direction constraints and spatial accessibility constraints of robots; this makes the path planning of dual-robot more complex and difficult to solve. Consequently, the off-line path planning method based on manual is difficult to meet the needs of actual production; it is urgent to study a kind of dual-robot automatic path planning algorithm for welding tasks of large complex components.

Some research in recent years has focused on the path planning problems for robot welding, it can be roughly divided into two aspects: spot welding and arc welding (Zhou et al., 2022). Generally, they were modelled as traveling salesman or multi traveling salesman problems, and then solved by using heuristic search algorithm (Wang et al., 2015). For the path planning problem of spot welding robot, its welding constraints are relatively weak, for example, there are no constraints related to welding direction. Zhang et al. (2019) discussed a motion planning method of spot welding industrial robot. Aiming at the coordination task of multi robot spot welding on the side of BIW, Hou et al. (2017) studied a welding path planning algorithm based on ant colony particle swarm optimization; the results showed that the algorithm can effectively improve the welding efficiency. Wang et al. (2017) modelled the path planning problem of spot welding robot as a traveling salesman problem; and a levy PSO algorithm was proposed to solve the optimal solution of welding path through the study of discrete Levy flight and discrete PSO algorithm. For the path planning problem of arc welding robot, it is more complex and difficult to solve because it has many constraints. Chen et al. (2016) proposed a path-planning algorithm using image processing for three-dimensional arc welding of the automatic welding robot system. Zhou et al. (2021) presented a point cloud based robotic system for arc welding, and introduced several methods for effective welding path planning. Wang et al. (2015) took the manual hydraulic transport frame as the research object, transformed MTSP into TSP by introducing virtual points, and the artificial bee colony algorithm based on state transition strategy was used to solve the global optimal welding path.

At present, although there are some research results on the path planning of multi robot coordinated welding, it is mainly focused on the spot welding robot, and the research on the path planning of arc welding robot is relatively few. Moreover, many existing studies only consider a single objective, or convert multi-objective problem into a single objective optimization problem for solution, and the consideration of the problem is not comprehensive. In addition, large complex components have many welding seams and complex constraints, the existing algorithms cannot always guarantee the efficiency and accuracy. It is still necessary to study a dual-robot welding path planning algorithm with high efficiency and strong robustness.

Motivated by the aforementioned discussion, the purpose of this paper is to propose a dual-robot cooperative arc welding path planning algorithm based on NSGA-II (Deb et al., 2002) with faster convergence rate and stronger robustness. The main contribution of the work is summarized as follows:

- (1) Aiming at the dual-robot arc welding problem of large complex components, considering multiple constraints and objectives, a multi-variable, multi-constraint and multi-objective optimization model is established.
- (2) An improved multi-objective genetic algorithm (R-NSGA-II) is proposed to solve this optimization problem, it introduces a recombination mechanism to replace the mutation operation of NSGA-II.
- (3) The R-NSGA-II is used to solve the welding path planning problem of dual robots, similarly, the NSGA-II and OMOPSO (Babu et al., 2007) are used for comparison.

The article consists of five sections including introduction (Section 1). Section 2 establishes the optimization model of dual-robot welding. Section 3 proposes the R-NSGA-II algorithm and it is used to solve the welding path planning problem of dual robots in Section 4. Section 5 briefly describes the obtained results and future works.

2. PATH PLANNING MODELING

This section introduces the description and modeling of the dual-robot welding problem for large and complex components.

2.1 Problem description

For the path planning problem of spot welding robot, if the collision between robots is not considered, it can be simplified as a traveling salesman problem. In the traditional traveling salesman problem, the city to be visited can often be simplified as a point, and the route between two adjacent cities is definite and unique. However, the welding seam has practical significance and cannot be simplified into a single point in the dual-robot cooperative welding problem. In addition, there are some complex constraints between the welding seams, and there are many cooperative and competitive relationships between the two robots. Therefore, the traditional traveling salesman model is not suitable for the traversal of the welding seams.

The three-dimensional schematic diagram of a large complex component is shown in Fig. 1. The welding path combination

of dual-robot welding system is a complex problem for large complex components, it includes the relationship between cooperation and competition of robots. For example, some welds require two robots to weld synchronously to prevent deformation; some welding seams are located in the common accessible area of the two robots, and the robots compete for welding according to the optimization objectives; some seams cannot be welded at the same time due to process requirements, and the welding time of each other needs to be staggered. Therefore, it is necessary to establish a more reasonable task planning model to realize the cooperative welding of the two robots, and to ensure a shorter welding time and welding path. It can be seen that this problem is a multi-variable and multi-constraint optimization problem. The welding area is divided into the exclusive welding area and the competitive welding area of each robot, the division of welding area is shown in Fig. 2. A dual robot cooperative welding path planning model is established by extracting information such as the ID, length, type, starting point and ending point posture of the welding seam.

2.2 Variable selection and definition

The selection of variables is the first step of path planning modeling, the known constants and welding variables are defined as follows.

2.2.1 Definition of known constants

First, the welding seam set is defined as follows:

$$\mathbf{W} = [1, 2, 3, \dots, m], \quad (1)$$

where the number in brackets represents the ID of each seam, the corresponding starting point coordinate set is

$$\mathbf{S} = [s_1, s_2, \dots, s_m], \quad (2)$$

and the ending point coordinate set is

$$\mathbf{E} = [e_1, e_2, \dots, e_m], \quad (3)$$

each point contains three coordinates: x, y and z. The length of the welding seam is a constant value, defined as:

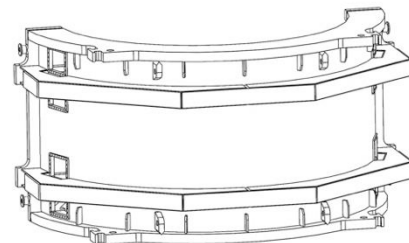


Fig. 1. Schematic diagram of a large complex component.

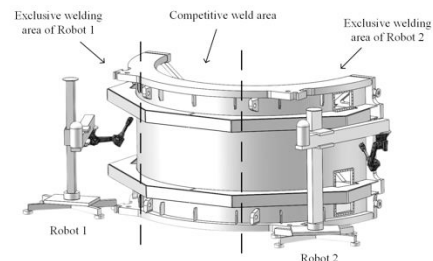


Fig. 2. Schematic diagram of welding area division

$$\mathbf{l} = [l_1, l_2, l_3, \dots, l_m], \quad (4)$$

where l_m is the length of welding seam m . In addition, the robot set is defined as $r = [1, 2]$, and the speed of the robot is defined as:

$$\mathbf{v} = [v_1, v_2], \quad (5)$$

where v_1 and v_2 are the welding speed and no-load speed of the robot respectively.

2.2.2 Definition of robot variables

The exclusive welding seam set of robots 1 and 2 are defined as $\mathbf{w}_1 = [w_{11}, w_{12}, \dots, w_{1m_1}]$, $\mathbf{w}_2 = [w_{21}, w_{22}, \dots, w_{2m_2}]$ respectively, where m_1 and m_2 are the exclusive welding seam numbers of robot 1 and robot 2 respectively. In addition, the competitive welding seam set is defined as $\mathbf{s} = [s_1, s_2, \dots, s_n]$, where n is the number of competitive seams. After task allocation, the final welding sequence obtained by the two robots are $\mathbf{p}_1 = [p_{11}, p_{12}, \dots, p_{1o_1}]$ and $\mathbf{p}_2 = [p_{21}, p_{22}, \dots, p_{2o_2}]$, where o_1 and o_2 are the number of welding seams finally assigned to robot 1 and robot 2, and $m = o_1 + o_2$. For a welding seam w_i , its starting point is s_i , the time required for the robot j to reach the s_i is defined as $t_{i,1}^j$, and the time after welding all the seams is $t_{i,2}^j$, i. e. $\mathbf{t}_i^j = [t_{i,1}^j, t_{i,2}^j]$, hence the time for robot j to weld all welding seams is $T = t_{i,1}^j + t_{i,2}^j$, where $j \in [1, 2]$ and $\begin{cases} i \in [1, o_1], & \text{if } j=1 \\ i \in [1, o_2], & \text{if } j=2 \end{cases}$. The time for robot j to reach each seam is $\mathbf{t}^j = [t_{1,1}^j, t_{2,1}^j, \dots, t_{o_j,1}^j]$. When the welding is completed, the total waiting time of the robot is T_w , and the total no-load distance is D_{nd} .

2.3 Optimization objectives and constraints

There will be welding waiting time between the two robots when some welding seams need synchronous or asynchronous welding. Higher welding efficiency expects shorter waiting time and shorter no-load distance. Consequently, the non-load distance and waiting time of the dual-robot system in the welding process are selected as the optimization objectives. According to the intermediate variables of the dual robot in the welding process, the following equations are derived:

$$D_{nd} = \sum_{j=1}^2 \sum_{i=1}^{o_j-1} \sqrt{(e_{p_{j(i)}}(1) - s_{p_{j(i+1)}}(1))^2 + (e_{p_{j(i)}}(2) - s_{p_{j(i+1)}}(2))^2 + (e_{p_{j(i)}}(3) - s_{p_{j(i+1)}}(3))^2} \quad (6)$$

waiting time:

$$T_w = (T_1 - new - T_1) + (T_2 - new - T_2) \quad (7)$$

where T_1 and T_2 are the welding time of robots 1 and 2 without considering various constraints, correspondingly, $T_1 - new$ and $T_2 - new$ are the actual welding time of robots 1 and 2 after considering various constraints such as synchronous welding, welding direction and collision constraints.

The following constraints are included:

(1) Synchronous welding constraint ϕ

For two simultaneous welding seams w_i and w_j , it is necessary to meet:

$$\alpha_i = \alpha_j, \beta_i = \beta_j, \quad (8)$$

where $w_i \in \mathbf{w}_1, w_j \in \mathbf{w}_2$, α_i and β_i are the start and end times of welding seam w_i , respectively.

(2) Welding direction constraint ϕ_2

Some welding seams have a fixed welding direction:

$$d_j = \begin{cases} 0, & \text{forward welding} \\ 1, & \text{reverse welding} \end{cases}, j \in W \quad (9)$$

(3) Collision constraint ϕ_3

$$R_i(q_i) \cap R_j(q_j) = \emptyset, \quad (10)$$

where q_i is the joint vector of robot i , $i, j \in r$, and $R_i(q_i)$ is the space occupied by the robot in this pose.

2.4 Path planning problem modelling

Therefore, the multi-objective dual-robot cooperative arc welding path planning model is established as follows:

$$\begin{aligned} \min F &= [D_{nd}, T_w] \\ \text{s.t.} \begin{cases} \phi_1 : \alpha_i = \alpha_j, \beta_i = \beta_j & (i, j \in W) \\ \phi_2 : d_j = \begin{cases} 0, & \text{forward welding} \\ 1, & \text{reverse welding} \end{cases} & (j \in W) \\ \phi_3 : R_i(q_i) \cap R_j(q_j) = \emptyset & (i, j \in r) \end{cases} \end{aligned} \quad (11)$$

3. THE PROPOSED PATH PLANNING ALGORITHM

This section introduces the proposed dual robot welding path planning algorithm (R-NSGA-II), including the main process and algorithm details. This method is proposed based on NSGA-II, and improved it according to the characteristics of dual robot welding problem. The diversity among individuals is increased by introducing recombination operator to replace the previous mutation operator, and the algorithm can converge to the global optimal solution more accurately.

3.1 Main procedure of the proposed algorithm

The main procedure of the proposed algorithm is described in Algorithm 1. First, population $\mathbf{P}_0$ with N individuals is initialized (line 1). Then the main loop started until the termination criteria are satisfied. The objective values of all individuals are calculated (line 4), and the non-dominated solution set $\mathbf{A}_t$ is generated by using fast non-dominated sorting method (line 5), and they are regarded as the parent population. Next, N offspring individuals are generated through crossover and recombination operations (lines 6~7). In order to preserve the excellent characteristics in the process of evolution, the offspring and parent populations are combined into population $\mathbf{Q}_{t+1}$ (line 8), and the best N individuals are selected as the new population $\mathbf{P}_{t+1}$ (line 9). A cycle ends here. When the algorithm reached the termination criteria, approximate pareto set $\mathbf{A}_t$ is output as the welding motion path of dual-robot.

Algorithm 1: Main procedure of the proposed algorithm

Input: Set $\mathbf{W}$, start point set $\mathbf{S}$, end point set $\mathbf{E}$, length set $\mathbf{l}$, direction set $\mathbf{d}$, robot speed $\mathbf{v}$, population size N

Output: approximate pareto set $\mathbf{A}_t$

- 1 $\mathbf{P}_0 \leftarrow$ Generate an initial population
- 2 $t \leftarrow 1$
- 3 **while** termination criteria are not satisfied **do**
- 4 $\mathbf{F}_t \leftarrow$ Calculate fitness values for all individuals
- 5 $\mathbf{A}_t \leftarrow$ Generate non-dominated solution set from $\mathbf{P}_t$
- 6 $\mathbf{Q}_{t+1}^1 \leftarrow$ Call crossover algorithm to generate N individuals
- 7 $\mathbf{Q}_{t+1}^2 \leftarrow$ Call recombination algorithm
- 8 $\mathbf{Q}_{t+1} \leftarrow \mathbf{P}_t \cup \mathbf{Q}_{t+1}^1 \cup \mathbf{Q}_{t+1}^2$
- 9 $\mathbf{P}_{t+1} \leftarrow$ Select the best N individuals
- 10 $t \leftarrow t + 1$
- 11 **end while**
- 12 **Return** $\mathbf{A}_t$

3.2 Initialization

The purpose of initialization is to generate an initialized population, and provide an initial solution for population evolution. An individual is generated with the ID of the welding seam as the variable as follows:

$$\mathbf{p} = [p_{11}, p_{12}, \dots, p_{1m_1}, p_{21}, p_{22}, \dots, p_{2m_2}, s_1, s_2, \dots, s_n] \quad (12)$$

where $p_i \in \mathbf{W}$, $p_{1m_1} \in \mathbf{w}_1$, $p_{2m_2} \in \mathbf{w}_2$, and $s_n \in \mathbf{s}$, and they do not repeat, i. e. $m_1 + m_2 + n = m$, $i \in [1, m]$. The competitive welding seams $\mathbf{s} = [s_1, s_2, \dots, s_n]$ are located in the common workspace of two robots, so they are necessary to compete for welding according to the optimization objectives. In the initialization phase, $s_n \in \mathbf{s}$ is randomly assigned to robot 1 or robot 2. Finally, the initial population $\mathbf{P}_0 = [\mathbf{p}_1, \mathbf{p}_2, \dots, \mathbf{p}_N]$ with size N is generated.

3.3 Fitness assignment and individual selection

In order to compare the advantages and disadvantages of two individuals, it is necessary to assign appropriate fitness values to them. The fitness assignment algorithm is list in Algorithm 2. At iteration t , firstly, the function values of each individual in $\mathbf{P}_t$ are calculated according to formulas (6) and (7) (lines 1~2). Then, $\mathbf{P}_t$ can be divided into l levels by using the fast non-dominated sorting algorithm (Deb et al., 2002) (line 3), and the individuals in the first level are stored in $\mathbf{A}_t$ as the non-dominated solution set (line 4). In addition, the individuals in $\mathbf{A}_t$ are also regarded as elite individuals to generate the next generation of population. Next, two fitness values are selected: one is the number of levels l (line 6), because the individuals in level l can dominate the individuals in level $(l+1)$. And the other is the crowding distance of each individual in the population (line 7), the individual with greater crowding distance is preferred for the individual selection on the same level. Finally, a fitness matrix $\mathbf{F}_t$ and an elite solution set $\mathbf{A}_t$ are output.

Algorithm 2: Fitness assignment algorithm

Input: Current population $\mathbf{P}_t$ with size N

Output: Fitness matrix $\mathbf{F}_t$, archive set $\mathbf{A}_t$

- 1 $\mathbf{D}_n' \leftarrow$ Calculate the non-load distance of each individual
- 2 $T_w' \leftarrow$ Calculate the waiting time of each individual
- 3 $\mathbf{P}_t$ is divided into l levels by using fast non-dominated sorting
- 4 $\mathbf{A}_t \leftarrow$ Individuals in the first level as non-dominated solutions
- 5 **for** $i \leftarrow 1$ **to** N **do**
- 6 $\mathbf{F}_t^{i,1} \leftarrow l$, the number of the individuals on l -th level
- 7 $\mathbf{F}_t^{i,2} \leftarrow$ The crowding distance of the individual on l -th level
- 8 **end for**
- 9 **Return** $\mathbf{F}_t$, $\mathbf{A}_t$

3.4 Crossover operation

The offspring population $\mathbf{Q}_{t+1}^1$ is generated through individual crossover operation when elite solution set $\mathbf{A}_t$ is selected. The specific algorithm is listed in Algorithm 3. The size of archive $\mathbf{A}_t$ is N' , and M individuals are randomly selected as the parent population for cross operation (line 2), and the rest as the mother population (line 3). The algorithm sets the crossover probability γ , randomly selects the crossover point p_c (line 1), and performs the crossover operation (lines 6~7). After removing the repeated variables (lines 8~9), a new offspring population $\mathbf{Q}_{t+1}^1$ is generated.

3.5 Recombination operation

Cross operation is to generate new children by exchanging information between different individuals. The mutation operation of NSGA-II is to randomly change some gene values of some individuals within a certain range, this still cannot effectively realize the information exchange within individuals.

Algorithm 3: Crossover operation

Input: Elite archive $\mathbf{A}_t$ with size N' , welding seam set $\mathbf{W}$ with size m , crossover probability γ

Output: Offspring population $\mathbf{Q}_{t+1}^1$

- 1 $p_c \leftarrow \lceil (m-1) \times \text{rand} \rceil$, ($\lceil \cdot \rceil$ means rounding up)
- 2 $\mathbf{P}_t^f \leftarrow$ Randomly select $N'/2$ individuals from $\mathbf{A}_t$
- 3 $\mathbf{P}_t^m \leftarrow \mathbf{A}_t - \mathbf{P}_t^f$
- 4 $r_1 \leftarrow$ Generate a random number of $[0,1]$
- 5 **if** $r_1 < \gamma$ **then**
- 6 $\mathbf{Q}_{t+1}^f \leftarrow [\mathbf{P}_t^f((\gamma_c + 1):m), \mathbf{P}_t^m]$
- 7 $\mathbf{Q}_{t+1}^m \leftarrow [\mathbf{P}_t^m((\gamma_c + 1):m), \mathbf{P}_t^f]$
- 8 $\mathbf{Q}_{t+1}^f \leftarrow$ Remove duplicate variables
- 9 $\mathbf{Q}_{t+1}^m \leftarrow$ Remove duplicate variables
- 10 **end if**
- 11 $\mathbf{Q}_{t+1}^1 \leftarrow \mathbf{Q}_{t+1}^f \cup \mathbf{Q}_{t+1}^m$
- 12 **Return** $\mathbf{Q}_{t+1}^1$

Moreover, welding ID is a discrete integer variable and cannot be repeated with each other, the variation operation cannot meet the requirements. Therefore, this paper proposes a recombination operation, which can realize the information interaction within individuals while satisfying the mutation operation, and increase the diversity of individuals and population.

The recombination algorithm is listed in Algorithm 4. $\lambda_2 \times N'$ individuals are selected for recombination operation by setting recombination probability λ_1 . Firstly, m integers are randomly arranged and stored in $\tilde{\mathbf{r}}_1$ and $\tilde{\mathbf{r}}_2$ averagely (lines 3–5). Then, for the individuals to be recombined in the population $\mathbf{Q}_{t+1}^1$, the variables at $\tilde{\mathbf{r}}_1$ and $\tilde{\mathbf{r}}_2$ are exchanged to complete the individual reorganization (lines 6–10). Finally, $\mathbf{Q}_{t+1}^2$ is exported as the recombined population.

Algorithm 4: Recombination algorithm

Input: Population $\mathbf{Q}_{t+1}^1$ with size N' , recombination rate λ_1 , recombination ratio λ_2 , number of variables m

Output: Offspring population $\mathbf{Q}_{t+1}^2$

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1   $r_2 \leftarrow$  Generate a random number from  $[0,1]$ 
2  if  $r_2 < \lambda_1$  then
3     $\tilde{\mathbf{r}} \leftarrow$  Randomly generate  $m$  different integers from  $[1,m]$ 
4     $\tilde{\mathbf{r}}_1 \leftarrow$  Randomly select  $m/2$  elements from  $\tilde{\mathbf{r}}$ 
5     $\tilde{\mathbf{r}}_2 \leftarrow \tilde{\mathbf{r}} - \tilde{\mathbf{r}}_1$ 
6     $\mathbf{Q}_{t+1}^2 \leftarrow \mathbf{Q}_{t+1}^1$ 
7    for  $i \leftarrow 1$  to  $N' \times \lambda_2$  do
8      for  $j \leftarrow 1$  to  $m/2$  do
9         $\mathbf{Q}_{t+1}^2(i, \tilde{\mathbf{r}}_1(j)) \leftarrow \mathbf{Q}_{t+1}^1(i, \tilde{\mathbf{r}}_2(j))$ 
10        $\mathbf{Q}_{t+1}^2(i, \tilde{\mathbf{r}}_2(j)) \leftarrow \mathbf{Q}_{t+1}^1(i, \tilde{\mathbf{r}}_1(j))$ 
12     end for
13   end for
14 end if
15 Return  $\mathbf{Q}_{t+1}^2$ 

```

3.6 Elite individual retention strategy

In order to retain the excellent characteristics in the process of evolution, the parent and the offspring population are combined, and then individual selection is carried out again. That is $\mathbf{Q}_{t+1} = \mathbf{P}_t \cup \mathbf{Q}_{t+1}^1 \cup \mathbf{Q}_{t+1}^2$. Obviously, the size of $\mathbf{Q}_{t+1}$ must be greater than N , and N individuals need to be selected as the new population $\mathbf{P}_{t+1}$ for the next iteration.

3.7 Termination criterion

Generally, the maximum number of generations or objective function evaluations is set as the termination criterion. The algorithm terminates after the termination criterion is satisfied, and the individuals saved in $\mathbf{A}_t$ are output as the approximate optimal solution set.

4. EXPERIMENTS AND RESULTS ANALYSIS

In this section, a series of experiments are performed to evaluate the proposed R-NSGA-II algorithm in solving the

dual-robot cooperative welding path planning problem. Its performance is mainly evaluated by Pareto results, welding time and welding path. In addition, NSGA-II and OMOPSO are selected for comparison.

4.1 Typical welding seam selection and parameter setting

Several typical welding seams of large and complex components shown in Fig. 1 are selected for experiment, and the specific results are shown in Fig. 3. The details of these seams are listed in Table 1.

The R-NSGA-II, NSGA-II and OMOPSO are selected as competitors in the path planning experiment of these typical welding seams. The detailed parameters of R-NSGA-II mentioned in Section 3 are listed in Table 2. The above algorithms and experiments are implemented in MATLAB 2021b environment. The hardware environment is Intel i9@2.3 GHz CPU and 32G memory.

4.2 Comparison of solution distribution of three algorithms

The solution distribution diagrams of R-NSGA-II, NSGA-II and OMOPSO are listed in Fig. 4 through a series of experiments. It can be seen from the figure that the solutions obtained by using R-NSGA-II can completely dominate the solutions obtained by applying the other two algorithms. It has obvious advantages in both waiting time and no-load distance. This shows that the R-NSGA-II algorithm has a better effect on the dual-robot arc welding path planning problem. A solution $[7.23, 5.37 \times 10^3]$ is selected for dual-robot path planning, and the solution of NSGA-II at the same location is $[12.55, 6.00 \times 10^3]$. It can be obtained that the R-NSGA-II reduces the waiting time by 42.39% and the no-load distance by 10.50% by introducing the recombination strategy.

4.3 Welding path planning results of dual-robot

According to the selected solution $[7.23, 5.37 \times 10^3]$, that is the waiting time is 7.23 seconds, and the non-load distance is 5.37×10^3 millimeters. The path of robot 1 is $path_1 = 17 \rightarrow 18 \rightarrow 19 \rightarrow 16 \rightarrow 6 \rightarrow 13 \rightarrow 5 \rightarrow 4 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 15 \rightarrow 26 \rightarrow 14$, and the path of robot 2 is $path_2 = 12 \rightarrow 22 \rightarrow 25 \rightarrow 24 \rightarrow 23 \rightarrow 20 \rightarrow 21 \rightarrow 27 \rightarrow 11 \rightarrow 10 \rightarrow 7 \rightarrow 8 \rightarrow 9$. The competitive welding seam 13 is assigned to robot 1, and the specific results are shown in Fig. 5. In addition, robot 2 waits for robot 1 for 7.23 seconds at the position of welding seam 7, and then they work together to complete seams 3 and 7.

5. CONCLUSIONS

In this paper, a novel improved multi-objective genetic algorithm (R-NSGA-II) is proposed by introducing a recombination strategy. The purpose is to solve the problem of collaborative welding path planning of dual-robot for large complex components. To verify the effectiveness of R-NSGA-II algorithm, 27 typical welding seams are selected from a large complex component, and a series of experiments are carried out by using the proposed algorithm and NSGA-II, OMOPSO. The simulation show R-NSGA-II can get better pareto front and better path planning effect. Particularly, the

R-NSGA-II can reduce 42.39% of waiting time and 10.50% of no-load distance compared with NSGA-II.

However, the algorithm proposed in this paper still needs further research. Firstly, there are more optimization objectives need to be considered, for example, the balance of welding time. Secondly, whether the existing various adjustment parameters are appropriate needs to be further studied. Thirdly, the effect of the algorithm on more complex problems also needs to be tested. Therefore, a series of works need to be continued in the future to solve the above problems.

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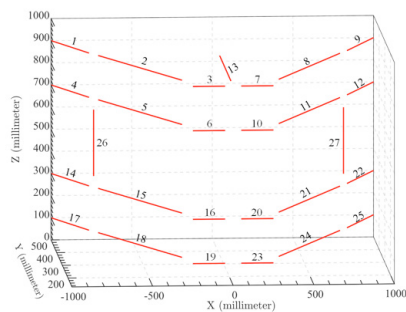


Fig. 3. Typical welds of large and complex components.

Table 1. Details of typical welds.

Parameters	Welding ID
Exclusive welds of robot 1	1~6, 14~19, 26
Exclusive welds of robot 2	7~12, 20~25, 27
Synchronous welds	3 and 7
Asynchronous welds	26 and 27
Competitive weld	13

Table 2. The parameters setting for R-NSGA-II.

Parameters	Values
Population size	$N = 100$
Welding speed of robot	$v_1 = 10 \text{ mm/s}$
No-load speed of robot	$v_2 = 20 \text{ mm/s}$
Crossover probability	$\gamma = 0.7$
Recombination probability	$\lambda_1 = 0.3$
Recombination ratio	$\lambda_2 = 0.5$

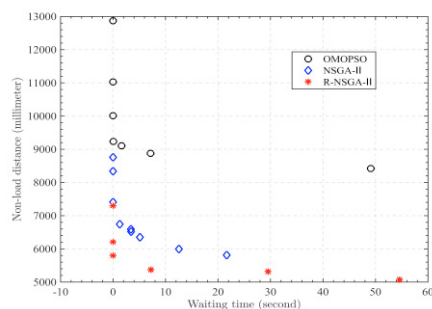


Fig. 4. Solution distribution diagrams of three algorithms

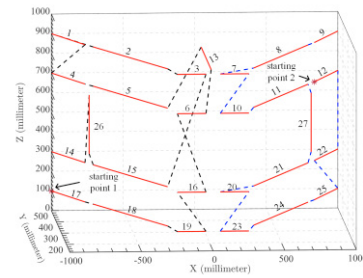


Fig. 5. Path planning results of dual-robot welding

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